

Jumpstarter ZPIRE

Team Name: TrailPulse

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Project Overview

Powered by an AI-integrated platform, our solution integrates open data on local trails (elevation, surface, distance) with real-time athlete metrics to deliver four key capabilities:

- (1) **Posture Optimization:** Using mobile phone cameras, the system detects and corrects running/standing posture, addressing form-related injuries at the source.
- (2) **Location-Specific Recommendations:** Suggests optimal trails (e.g., country park routes vs. urban promenades) based on terrain features, aligning with users' fitness levels and goals.
- (3) **Real-Time Adaptive Coaching:** Monitors pace, heart rate, and exertion to keep runners in optimal training zones, providing immediate audio/haptic feedback during runs.
- (4) **Route-Tailored Training Plans:** Generates structured programs for iconic local routes—whether preparing for Lantau Peak ascents or Bowen Road tempo runs—ensuring training aligns with terrain demands.

Multi-Dimensional Reflections & Action Plans

Topography

- Public trails are ubiquitous, varied (elevation, surface), and underutilized as real-time training environments.

Training technique

- Leverage trails' diversity to offer route-aware training (e.g., hill technique on maintain vs. tempo on roads).

Opportunity

- Combine open trail data + on-site posture analytics for location-specific coaching.

Problem Faced:

1. Many runners train on public trails without personalized posture feedback, increasing injury risk and limiting performance gains.
2. During the running on the hiking track, some injuries or off-track may happen.

Solution Proposed:

1. Providing AI analysis app, in order to detect user's standing and running posture.
2. Deploy edge-enabled cameras to capture anonymized pose keypoints on trails; analyze form with AI; deliver real-time and post-run corrective guidance in a professional app. Once any emergency situation happened, the app will send alert to our data center.

Why this matters in HK?



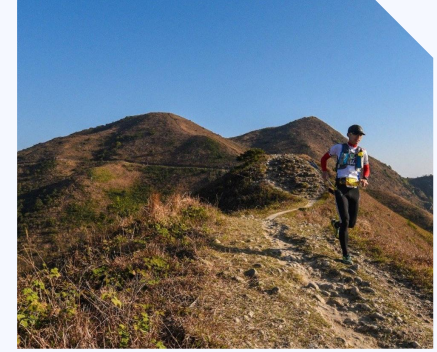
Many Running Activity

- High participation in outdoor running across Hong Kong's trails and promenades.



Injuries Caused

- Many runners use generic plans; on-site posture issues (overstriding, knee valgus, forward lean) go undetected.



Expensive Expenses

- Limited affordable, scalable options for personalized, real-time running-form coaching.

Who do we serve?

- **Primary:** Recreational and semi-competitive runners aged ~18–45 who train on Hong Kong trails and want performance gains or injury reduction.
- **Secondary:** University athletes, running clubs, campus sports departments, municipal recreation units.



What we deliver?

- 01 Posture** — Personalized, route-aware posture coaching delivered where runners train.
- 02 Emergency Support** — Real-time alerts and short corrective drills for immediate behavior change.
- 03 Analysis** — Post-run insights and structured training plans aligned with sports science.
- 04 Public Awareness** — Scalable public health benefit: safer running population and reduced demand for acute care.

Core Feature

Software

Trails Recommendation

Recommend trails and sessions based on elevation, surface, and user goals.

Coaching

Adaptive coaching: in-run audio/haptic micro-coaching and immediate suggested drills in-app.

Post-Run Report

Weekly balance report: training load vs. recovery suggestions per ACSM-derived rules.

Hardware

Posture Detection

Edge keypoint capture: cameras + local inference extract skeletal keypoints (no raw video storage).

Posture Real-Time Adjustment

Real-time posture analysis: classification of common issues (overstride, forward lean, knee valgus, cadence anomalies).

Off-Track /Emergency

Deploying the edgeAI with trained posture detection model on running tracks (Jetson Nano, ESP32 etc)

Work Principles

Camera → Pose Model
(MediaPipe/TF Lite/Edge model)

Anonymize & Compress Keypoint Vectors;
do not store or transmit identifiable video.

Track user's running route
detect if there is any off track
or emergency situation

On-Device

Privacy

Safety



Cloud/App

User

Matching

Collected Data will be uploaded
to cloud and be shown in
database and user's app

App receives real-time create
post-run reports;
Opt-in data sharing for with their
running partners.

Runners can match with other users
who ran the same trail before in the
app and meet up for friends

AI & Modeling

01

Rule-based heuristics +
Lightweight pose classifiers
for common faults.

02

Supervised models trained on
labeled keypoint sequences
(transfer learning from open
datasets + local labeled samples).

03

Personalization Layer (user baseline
+ incremental learning to adapt
cues).

Tools/Tech:

- (1) MediaPipe/TensorFlow Lite for edge;**
- (2) PyTorch/TF for model development;**
- (3) ZPIRE premium tools for rapid prototyping.**

Edge Device & Hardware Deployment

01



Device Choices: Raspberry Pi 4 / Jetson Nano, strong TPU for local pose inference depending on throughput and latency needs.

02



Camera Placement Guidelines: visible signage, consent flow, non-face framing where possible, typical sampling window (5–10 m segments).

03



Maintenance & Ops: solar power options, remote update pipeline, periodic model updates and animal's damage preventing support

Privacy, Consent & Ethics

Data Minimization

Store only skeletal keypoints and derived metadata;
No raw video or face features.

Government

Encryption, retention limits, audit logs, partnership MOUs with municipal/campus authorities.

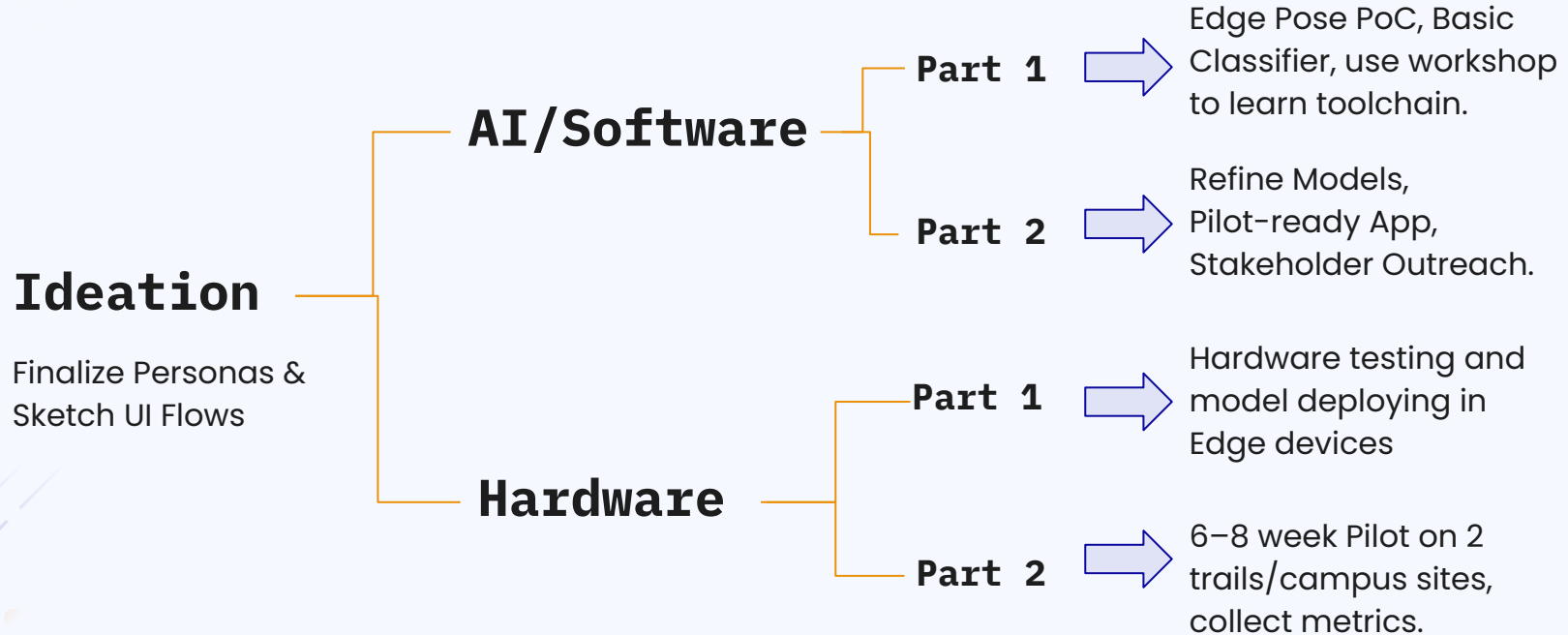
Consent & Signage

Clear trail signage (multilingual),
QR-based consent option,
Anonymous mode for passersby.

Human Safeguards

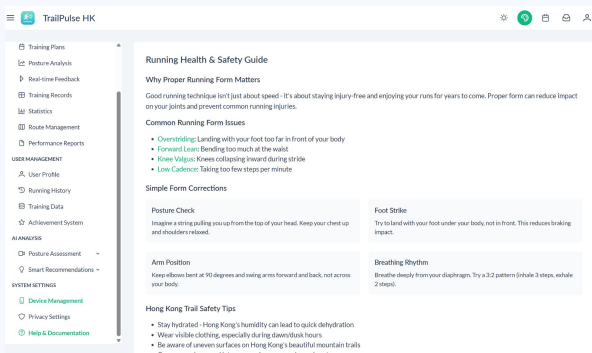
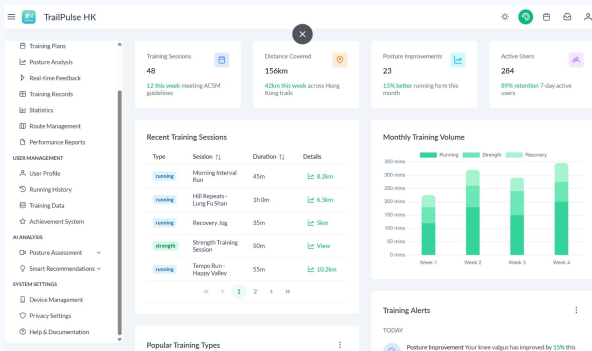
Human review & escalation pathways
if high-risk gait patterns detected.

Implementation Timeline



Welcome to Our Github

<https://github.com/3dsimon/trailpulse>



Resource

Database(from data.gov.hk):

- Hiking Trails in Country Parks:
<https://data.gov.hk/en-data/dataset/hk-afcd-afcdlist-hikingtrailscp>
- Feature Articles on Culture, Entertainment and Recreation – Sports and Related Activities in Hong Kong:
<https://data.gov.hk/en-data/dataset/hk-censtatd-tablechart-fa100281>
- Fitness Walking Tracks in 18 Districts:
<https://data.gov.hk/en-data/dataset/hk-lcsd-facility-facility-fw>
- Location of Jogging Tracks and Fitness Trails:
<https://data.gov.hk/sc-data/dataset/hk-lcsd-facility-facility-jtft>

Github:

- <https://github.com/3dsimon/trailpulse>

Data Visualization:

- <https://public.tableau.com/app/profile/tsz.hin.yuen/viz/HikingTrailsinCountryParks/1?publish=yes>
- <https://public.tableau.com/app/profile/tsz.hin.yuen/viz/GDPatmarketpricesofsportsandrelatedactivitiesinHongKongMn/1?publish=yes>

Thanks !

Do you have any questions?

Contact:

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This PPT is made by Yuhang ZHOU, ChunWa LUK, TszHin YUEN

